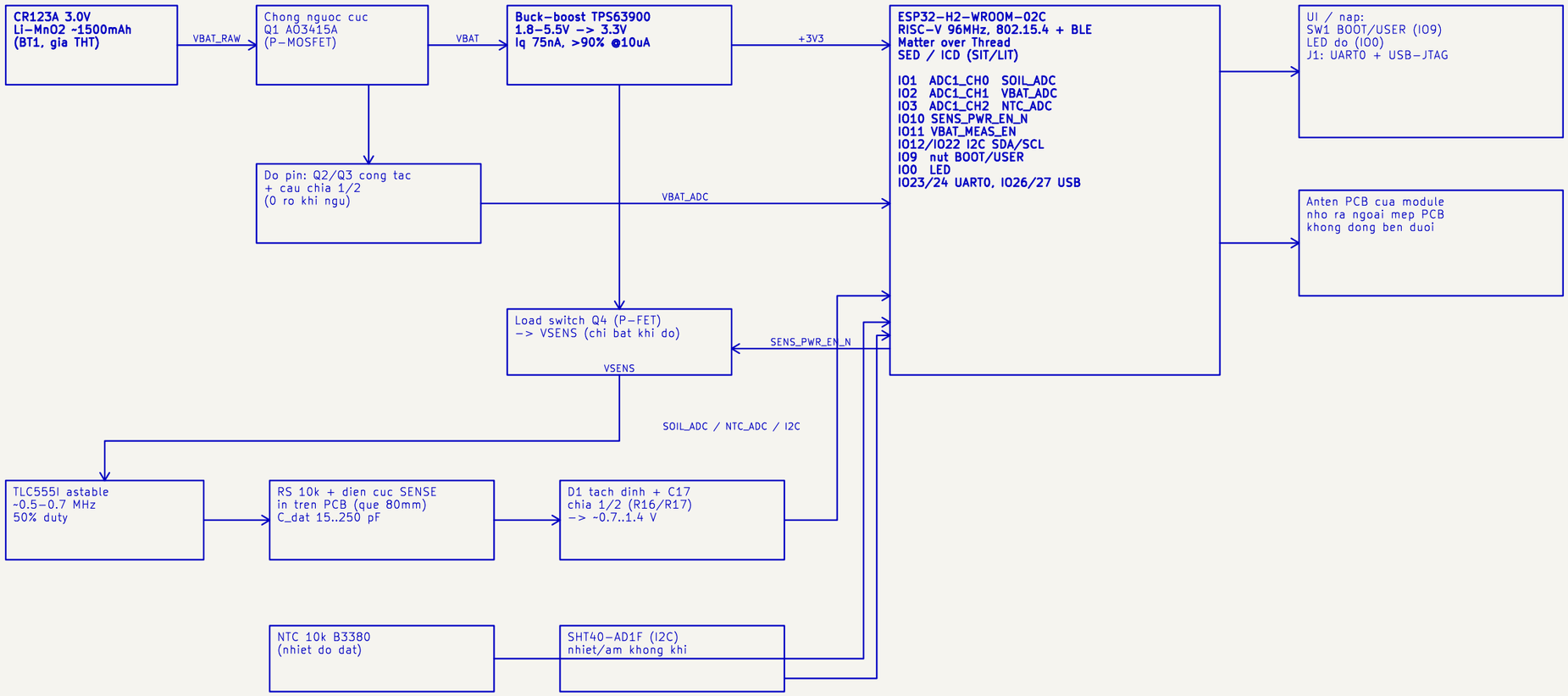
SO DO KHOI – H2 Soil Sensor (Matter over Thread, ESP32-H2, 1x CR123A)



Power

Power: CR123A,
RPP,
TPS63900 3V3,
VBAT sense,
VSENS

File: power.kicad_sch

MCU

MCU: ESP32-H2-WROOM-02C (Thread ICD)

File: mcu.kicad_sch

Sensors

Sensors: capacitive probe,
soil NTC,
SHT40

File: sensors.kicad_sch

UI

UI: button,
LED,
prog header,
test points

File: ui.kicad_sch

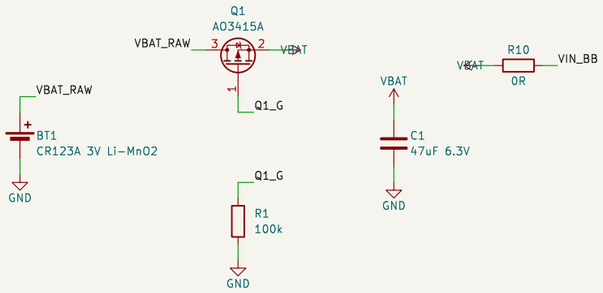
Root sheet: so do khoi + 4 sheet con (ket noi bang global label)
H2 Soil Sensor

Sheet: /
File: h2-soil-sensor.kicad_sch

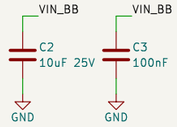
Title: H2 Soil Sensor – Matter over Thread (ESP32-H2, CR123A)

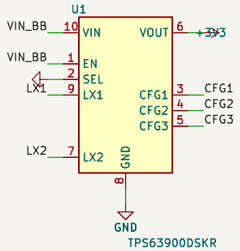
Size: A3	Date: 2026-11-11	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/5

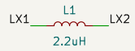
1) PIN + CHONG NGUOC CUC
Q1 P-MOSFET: D->pin(+), S->tai, G->GND qua R1.
Lap nguoc pin: Vgs=0, diode than phan cuc nguoc
-> khong co dong. R10 0R: cau do dong (PPK2).



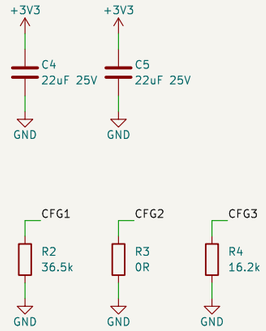
2) TPS63900 BUCK-BOOST -> 3.3V (Iq 75nA)
SEL=GND -> VO(1) theo CFG3=16.2k = 3.3V
CFG1=36.5k (VO(2)=3.3V), CFG2=0R: lin khong gioi han
R CFG 1%/100ppm, khong dat tu tren chan CFG.
L1 2.2uH Isat>=2A; COUT 2x22uF; CIN 10uF.

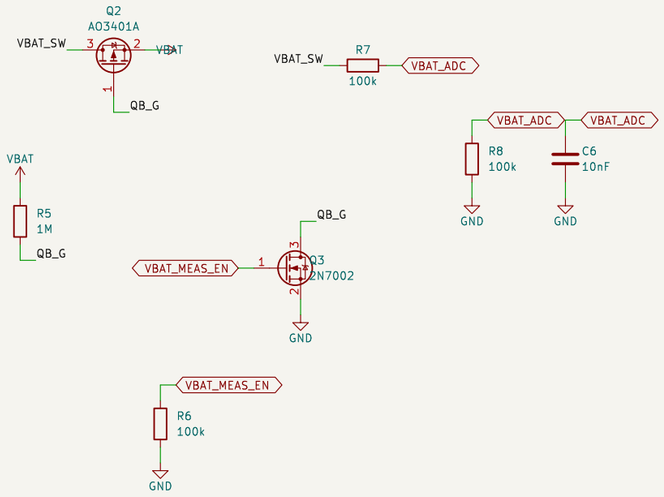




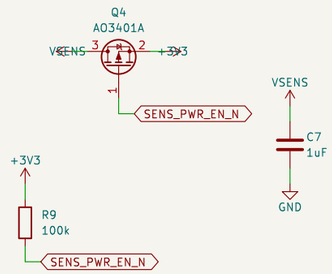


3) DO DIEN AP PIN (khong ro khi ngu)
VBAT_MEAS_EN=1 -> Q3 ON -> Q2 ON -> VBAT/2 -> IO2.
Khi ngu: R6 giu Q3 OFF, R5 giu Q2 OFF.

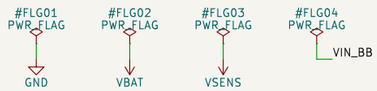




4) CONG TAC NGUON CAM BIEN VSENS
SENS_PWR_EN_N=0 -> Q4 ON: cap 3.3V cho TLC555,
NTC, SHT40, pull-up I2C. R9 giu Q4 OFF khi
GPIO Hi-Z (sleep/reset).

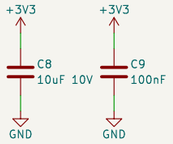


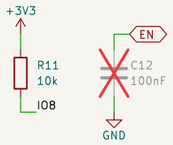
PWR_FLAG (cho ERC)

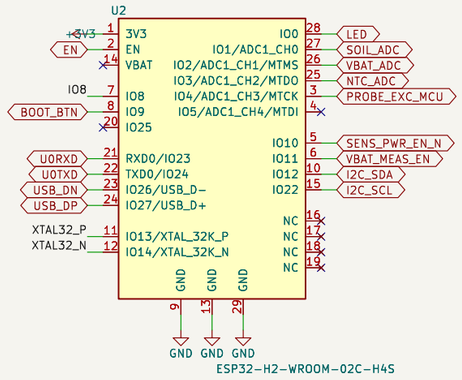


ESP32-H2-WROOM-02C, CR123A, TPS63900
H2 Soil Sensor – Matter over Thread
Sheet: /Power/
File: power.kicad_sch
Title: Power: CR123A, RPP, TPS63900 3V3, VBAT sense, VSENS
Size: A3 | Date: 2026-11-11 | **Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 2/5

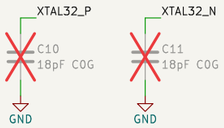
ESP32-H2-WROOM-02C (4MB: -H4S de co OTA). Vdd 3.0-3.6V.
Module da co RC tren EN (10k/1uF) -> C12 DNP.
Strapping: IO9=1 (R21, sheet UI) -> SPI boot; IO8=1 (R11).
IO10/IO11 la LP GPIO: giu muc khi light-sleep tat mien TOP.
ADC1: IO1=SOIL, IO2=VBAT, IO3=NTC. IO4: kich thich LEDC (tuy chon).









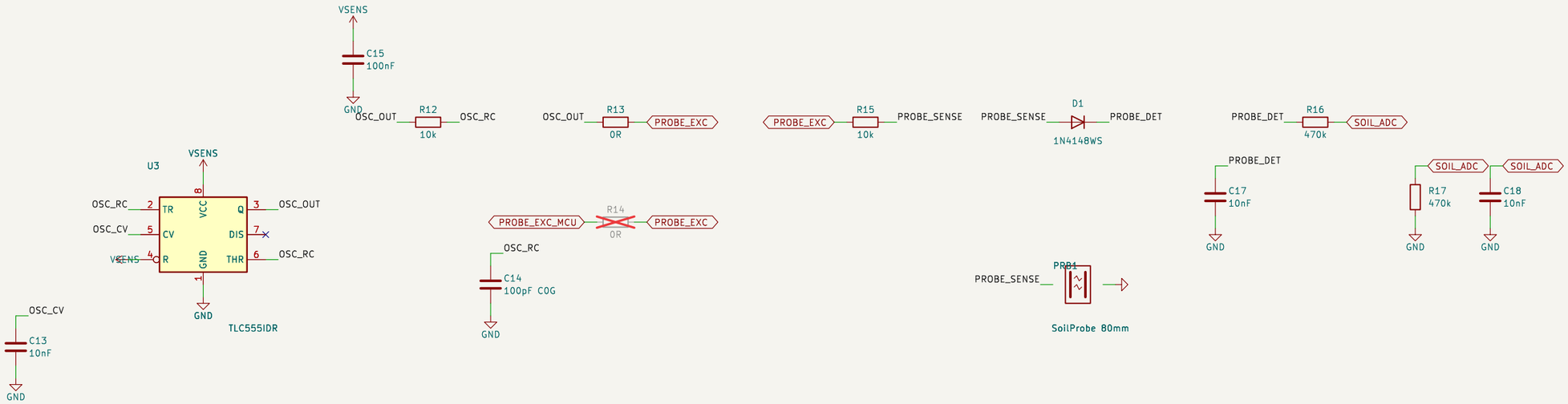


Anten PCB: module o mep tren cung, vung anten (6mm) nho ra ngoai PCB;
khong dong/GND/linh kien duoi va quanh anten; vo hop cach anten >=15mm.
Y1/C10/C11 (DNP): XTAL 32.768kHz tuy chon tren IO13/IO14.

ESP32-H2-WROOM-02C, CR123A, TPS63900
H2 Soil Sensor – Matter over Thread

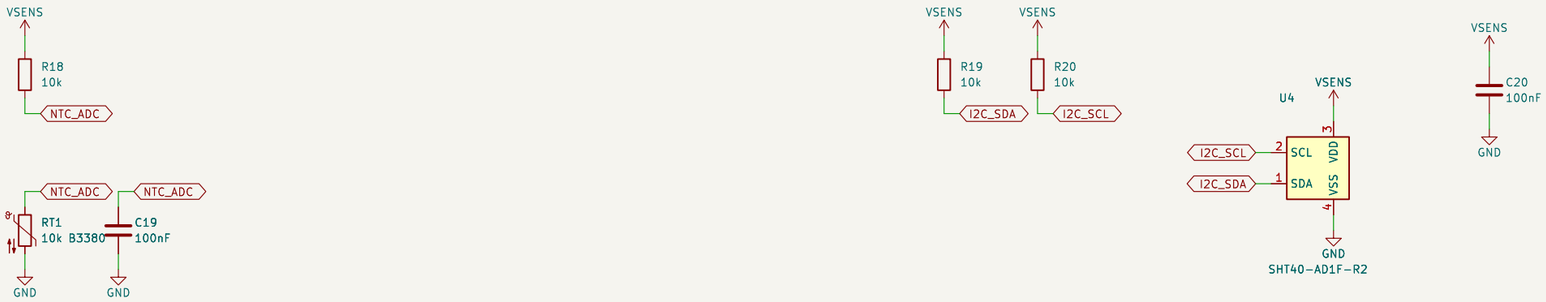
Sheet: /MCU/		
File: mcu.kicad_sch		
Title: MCU: ESP32-H2-WROOM-02C (Thread ICD)		
Size: A3	Date: 2026-11-11	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/5

DAU DO DIEN DUNG: TLC555 (ban I) dao dong ~0.5–0.7MHz, duty 50%
(RT=R12 10k, CT=C14 100pF C0G). OSC_OUT -> R13 -> RS(R15 10k) ->
dien cuc SENSE (C_dat ~15..250pF so voi GND) -> D1 tach dinh -> C17.
R16/R17 chia 1/2 vao dai ATTEN2 (0..1.9V). Dat cang uot -> C cang lon
-> bien do cang nho -> dien ap SOIL_ADC cang THAP.



NTC 10k B3380 (RT1) dat tren than que, ~5cm duoi mat dat:
nhiet do dat + bu nhiet cho dau do. Nuoi tu VSENS (0 ro khi ngu).

SHT40-AD1F (mang PTFE, IP67): nhiet/am khong khi.
Dat o vung co ranh phay cach nhiet, ho ra qua lo co mang.
Pull-up I2C noi VSENS: FW dat SDA/SCL Hi-Z truoc khi ngu.



ESP32-H2-WROOM-02C, CR123A, TPS63900

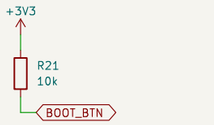
H2 Soil Sensor – Matter over Thread

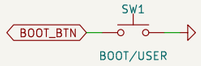
Sheet: /Sensors/
File: sensors.kicad_sch

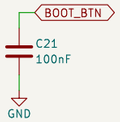
Title: Sensors: capacitive probe, soil NTC, SHT40

Size: A3	Date: 2026-11-11	**Rev: 1.0**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 4/5

SW1 (IO9/BOOT): 1 lan = do & bao cao ngay / mo commissioning:
2 lan = hieu chuan KHO; 3 lan = hieu chuan UOT; giu >=10s = factory
reset. Giu SW1 khi cap nguon/reset = che do nap (download).

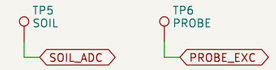




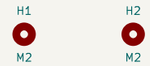


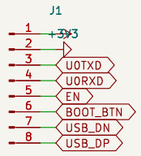












PROG

J1: 1=3V3 2=GND 3=TXD0 4=RXD0 5=EN 6=IO9 7=USB_D− 8=USB_D+
USB-Serial/JTAG: lap pin + noi D+/D−/GND (khong noi 3V3).
UART: thao pin, cap 3V3 tu adapter USB-UART 3.3V.

ESP32-H2-WROOM-02C, CR123A, TPS63900
H2 Soil Sensor – Matter over Thread
Sheet: /UI/
File: ui.kicad_sch
Title: UI: button, LED, prog header, test points
Size: A3 | Date: 2026-11-11 | **Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 5/5